Mechanics and Waves
Grado en Física
2025-2026

UNIVERSIDAD DE GRANADA

Experiment M9
Standing waves on a string

Victoria Torres Ochando
Martina Lodos Falcón

Laboratory session 30/04/2026

Abstract

This study aims to experimentally verify the frequency law governing transverse standing waves produced in a stretched string under fixed boundary conditions. The experimental methodology involves establishing standing wave patterns by varying the oscillation frequency for three distinct string tensions (0.80 N, 0.90 N, and 1.00 N), measuring the true resonant frequencies through stroboscopic synchronization. A linear regression framework based on the method of least squares ($y = ax$) is proposed to evaluate the direct proportionality between the frequency (f_n) and the vibrational mode (n), allowing the experimental determination of the string's linear mass density and the fundamental frequency (f_1). Additionally, the scope of the analysis includes assessing the potential presence of wave dispersion within the physical system by evaluating the relationship between angular frequency and the wave number.

Contents

1 Theoretical background **2**

2 Methodology and Materials **3**
2.1 Materials . . . 3
2.2 Methodology . . . 3

3 Results and discussion **4**
3.1 Experimental measurements . . . 4
3.2 Graphical representation of f and determination of ρ and f_1 . . . 4
3.3 Study of Linear Density through the Joint Relationship f_n/n vs $T^{1/2}$ . . . 5
3.4 Wave Analysis and Dispersion Relation . . . 7

4 Conclusion **7**

5 Appendix **8**
5.1 Graphics and fittings . . . 8
5.2 Data tables . . . 11
5.3 Uncertainties . . . 13
5.3.1 General Uncertainties Calculations . . . 13
5.3.2 Direct measurements . . . 14
5.3.3 Specific Uncertainty Calculations . . . 14

1 Theoretical background

The propagation of a one-dimensional wave along the x-direction is mathematically governed by the general wave equation:

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2} = 0, \tag{1}$$

where v represents the phase velocity of the wave. In the specific case of a transverse wave traveling through a flexible, ideal string under a constant tension T and possessing a linear mass density ρ, this propagation velocity is physically determined by the mechanical properties of the medium as follows.

$$v = \left(\frac{T}{\rho}\right)^{\frac{1}{2}} \tag{2}$$

A particular solution to the wave equation can be expressed using complex exponential notation as

$$\psi(x,t) = Ae^{i(\omega t - kx)}, \tag{3}$$

where A is the amplitude, $\omega = 2\pi f$ is the angular frequency, and $k = 2\pi/\lambda$ is the wave number.

When considering a bounded system, a general solution can be constructed by superimposing two traveling waves of identical frequency and amplitude moving in opposite directions along the string. For a string of length L fixed firmly at both ends ($x = 0$ and $x = L$), the physical boundaries restrict the displacement, imposing the following boundary conditions:

$$\psi(0,t) = 0 \quad \text{and} \quad \psi(L,t) = 0$$

Applying these boundary conditions to the superposition of the counter-propagating waves yields a standing wave equation:

$$\psi(x,t) = 2Asen(kx)sen(\omega t). \tag{4}$$

In this state, the resulting motion does not propagate along the spatial axis. Instead, it forms a stationary pattern where the local amplitude depends exclusively on the position, given by Amplitude $= 2A\, sen(kx)$. This spatial distribution produces alternating points of zero displacement known as nodes and points of maximum displacement known as vientres.

Consequently, stable standing waves can only transition into resonance when the physical length of the string accommodates an integer number of half-wavelengths:

$$L = n\frac{\lambda_n}{2} \quad \Longrightarrow \quad \lambda_n = \frac{2L}{n}, \tag{5}$$

where $n \in N$ denotes the vibrational mode. Given that the phase velocity relates the frequency and wavelength via $v = \lambda_n f_n$, substituting this boundary restriction into the velocity relationship provides the fundamental frequency law for a stretched string:

$$f_n = \frac{n}{2L}\left(\frac{T}{\rho}\right)^{\frac{1}{2}} \tag{6}$$

For $n = 1$, the system vibrates at its lowest resonance, known as the fundamental mode (f_1). Thus, the frequency of any higher-order harmonic mode (f_n) is a direct integer multiple of the fundamental frequency.

$$f_n = nf_1 \tag{7}$$

This linear relationship establishes the discrete frequency law to be verified experimentally for each applied tension. Finally, if the string departs from ideal behavior due to internal longitudinal elasticity, the system experiences wave dispersion. This phenomenon manifests as a

non-linear relationship between the angular frequency and the wave number, expressed through the dispersion relation:

$$\omega^2 = \frac{T}{\rho}k^2 + \alpha k^4, \tag{8}$$

where α is a characteristic dispersion coefficient. The validity of the ideal model versus the dispersive behavior can be evaluated by analyzing the linearity of ω^2 as a function of k^2.

2 Methodology and Materials

2.1 Materials

The laboratory setup consisted of the following equipment:

- A **string (model SE-8050)** with a theoretical linear mass density of $\rho_{theo} = 0,33$ g/m.
- A **mechanical wave driver** placed at one end, driven by a variable frequency generator.
- A **precision dynamometer** positioned at the opposite end to monitor the string tension T, featuring a sensitivity of $0,02$ N.
- A **calibrated metric rule** used to measure the effective length L from the vibrating rod to the pulley. The measured length was $L = 1,2450 \pm 0,00058$ m.
- A **digital stroboscope** used to determine the resonance frequencies by visual synchronization, with a sensitivity of 1 rpm.

2.2 Methodology

The steps followed in order to carry out the experiment are enumerated below:

1. **Initial Configuration:** With the generator powered off, the effective string length L was measured. The string tension was mechanically adjusted until the dynamometer stabilized at the first target value of $T = 1,00$ N.
2. **Normal Mode Detection:** The generator was turned on, and the frequency was swept around the 18–22 Hz range until the fundamental mode ($n = 1$) was established in the vertical plane. The amplitude was progressively optimized for higher-order modes to maintain stable nodes and antinodes.
3. **Stroboscopic Measurement:** To ensure the true resonance frequency was captured, the stroboscope sweep was started above 1000 rpm to avoid submultiples. The flashing rate was fine-tuned until the rapid movement of the string appeared entirely static due to synchronization. Then, the value in rpm was recorded.
4. **Statistical Replication:** For each mode, the generator amplitude was zeroed out and reset to perform at least 3 independent trials [**guion**]. The stroboscope was turned off between runs to prevent overheating.
5. **Harmonic and Tension Sweeps:** Steps 2 to 4 were repeated sequentially for higher harmonics ($n = 2, 3$, and 4). Finally, the entire procedure was replicated for the remaining tension levels of $T = 0,90$ N and $T = 0,80$ N.

3 Results and discussion

3.1 Experimental measurements

Table 7 details the experimental data collected during the laboratory session, while Table 1 summarizes the calculated average values for each vibrational mode. The indices a, b, and c denote the specific string tensions applied: $T_a = 1.000 \pm 0.012$ N, $T_b = 0.900 \pm 0.012$ N, and $T_c = 0.800 \pm 0.012$ N, respectively.

λ (m)	$\overline{f_a}$ (Hz)	$\overline{f_b}$ (Hz)	$\overline{f_c}$ (Hz)
2,4900±0,0023	22,094±0,011	21,678±0,011	21,411±0,011
1,2450±0,0012	42,239±0,011	42,867±0,011	41,600±0,011
0,83000±0,00077	66,328±0,011	65,778±0,011	61,661±0,011
0,62250±0,00058	89,984±0,011	87,317±0,011	84,178±0,011

Table 1: Wavelength λ and average frequencies $\overline{f_a}$, $\overline{f_b}$ and $\overline{f_c}$ for each vibrational mode, with their corresponding uncertainties.

3.2 Graphical representation of f and determination of ρ and f_1

To verify the frequency law for standing waves , a linear regression analysis was performed on the average frequency values against the vibrational mode n for each tension. Table 2 summarizes the resulting slopes, intercepts, and correlation coefficients, while the corresponding regression plots (Figures 2, 3 and 4) are provided in the Appendix for visual reference.

	a (Hz)	b (Hz)	R^2
Experiment A	$22,68 \pm 0,11$	$-1,4 \pm 1,5$	0,9992
Experiment B	$21,98 \pm 0,33$	$-0,55 \pm 0,91$	0,9998
Experiment C	$20,84 \pm 0,27$	$0,12 \pm 0,75$	0,9992

Table 2: Parameters for the linear regression of frequency versus vibrational mode for each string tension (experiment A, B and C)

The regression analysis yielded coefficients of determination ($R^2 > 0.999$) exceptionally close to unity, demonstrating an outstandingly strong linear relationship between f and n. Furthermore, although the theoretical model $f_n = nf_1$ suggests an intercept of zero, the regression analysis provided non-zero values for b. However, upon considering the calculated uncertainties, the value $b = 0$ falls within the confidence interval $[b - u(b), b + u(b)]$ for all cases. Consequently, the intercept can be considered statistically insignificant, validating the direct proportionality $y = ax$ as defined in the frequency law.

Based on these considerations, the linear density ρ of the string can be determined. By rearranging the theoretical expression for the frequency modes, we obtain the following relationship:

$$\rho = \frac{T}{4L^2a^2} \quad kg/m, \tag{9}$$

where T is the applied tension, L is the length of the string, and a is the slope obtained from the linear fit. The resulting values, converted to g/m to facilitate comparison with the reference data, are presented in Table 3.

ρ_a **g/m**	ρ_b **g/m**	ρ_c **g/m**	ρ **g/m**
0,3136 ± 0,0081	0,300 ± 0,012	$0,297 \pm 0,012$	$0,3037 \pm 0,0062$

Table 3: Density of the string for each experiment and average value ρ, with their corresponding errors.

As we can see, the density obtained experimentally is very close to the bibliographical value ($\rho_{bib.} = 0,33$ g/m, according to [1]), showing a relative error of $\epsilon = 7,965\%$.

Finally, the fundamental frequency (f_1) was evaluated. Table 4 compares the values measured directly during the laboratory session ($f_{1,e}$) with the theoretical values derived from the slopes of the linear fits ($f_{1,t} = a$). The high level of agreement between these two sets of values further reinforces the veracity of the experimental results.

	$f_{1,t}$ **(Hz)**	$f_{1,e}$ **(Hz)**	ϵ
Experiment A	$22,68 \pm 0,11$	$22,094 \pm 0,011$	2,65 %
Experiment B	$21,98 \pm 0,33$	$21,678 \pm 0,011$	1,38 %
Experiment C	$20,84 \pm 0,27$	$21,411 \pm 0,011$	2,67 %

Table 4: Experimental and theoretical frequency for the fundamental vibrational mode, $f_{1,e}$ and $f_{1,t}$, for each string tension

3.3 Study of Linear Density through the Joint Relationship f_n/n vs $T^{1/2}$

To obtain a more robust estimate of the string's linear density (ρ), a global analysis was performed by integrating all experimental data from the three different tension levels (T_a, T_b, T_c). According to the theoretical model:

$$\frac{f_n}{n} = \frac{1}{2L\sqrt{\rho}}\sqrt{T} \tag{10}$$

where the term $\frac{1}{2L\sqrt{\rho}}$ represents the slope (m) of a linear relationship between the average frequency-to-mode ratio and the square root of the tension.

The average values of $\overline{f_n/n}$ for each tension were calculated alongside their combined uncertainties. A linear regression forced through the origin ($y = mx$) was applied to the data points represented in Figure 1.

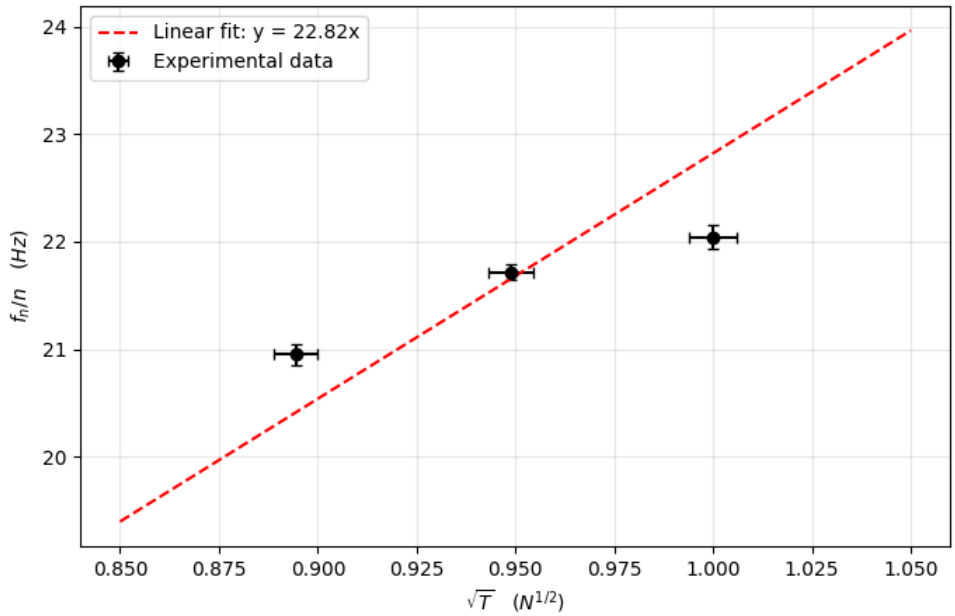


Figure 1: Global linear fit of $\overline{f_n/n}$ vs $\sqrt{T}$.

The results of the statistical analysis are summarized in Table 5.

Parameter	**Value**
Slope (m)	$22,822 \pm 0,053$ Hz/N$^{1/2}$
Coefficient of Determination (R^2)	$0,999350$
Reduced Chi-squared (χ^2_{red})	$41,2198$
Calculated Linear Density (ρ)	$\mathbf{0,3097 \pm 0,0015}$ g/m

Table 5: Global Fit Quality Report

The linear density obtained through the global method is $\rho_{global} = 0,3097 \pm 0,0015$ g/m. To evaluate the quality of this result, we compare it with the individual estimates from Section 2 and the theoretical value ($\rho_{th} = 0,33$ g/m) based on three criteria:

- **Goodness of Fit:** The global regression shows an $R^2 = 0,9994$, which indicates a higher degree of internal consistency compared to the individual fits. Although the reduced chi-squared value ($\chi^2_{red} \approx 41,2$) suggests that the experimental uncertainties might be slightly underestimated or that small systematic effects (like string stiffness) are present, the linear relationship remains remarkably solid across all tension regimes.

- **Precision:** The global method is significantly more **precise** than the individual measurements. By integrating the full dataset ($N = 36$), the uncertainty in the slope is minimized, resulting in a lower standard deviation for ρ. This demonstrates that the joint study effectively filters out random fluctuations (Type A errors) that affected the separate experiments in Section 2.

- **Accuracy:** In terms of **accuracy**, the result shows a consistent systematic deviation of approximately 6% below the theoretical value. Since this discrepancy is identical across all methods and exceeds the experimental uncertainty, it is likely not a measurement error but a systematic shift. This could be attributed to the nominal density being specified for a different state of the string or to a slight calibration bias in the dynamometer.

In conclusion, the global fit provides the most reliable determination of the linear density. It maintains the same accuracy as the individual fits while significantly enhancing the precision and statistical weight of the results.

3.4 Wave Analysis and Dispersion Relation

In this section, we analyze the wave behavior of the string by studying the dispersion relation. For an ideal string, the relationship between the angular frequency ($\omega = 2\pi f$) and the wave number ($k = n\pi/L$) is given by $\omega^2 = (T/\rho)k^2$. By representing ω^2 as a function of k^2, we can verify the linearity of the medium and determine the linear density ρ from the slope of the fit.

The results for the linear fits of ω^2 vs k^2 for each tension level are summarized in Table 6, and the resulting graphics in the apendix (figures 5, 6 and 7). The fits were performed using a linear model $y = mx$ forced through the origin.

Tension (N)	**Slope** m (m^2/s^2)	R^2	χ^2_{red}	ρ **(g/m)**
1.000	$3058,26 \pm 28,28$	0,998385	9493,32	$0.3270 \pm 0,0030$
0.900	$2961,26 \pm 17,52$	0,999444	3763,41	$0.3039 \pm 0,0018$
0.800	$2701,24 \pm 18,91$	0,998855	4803,76	$0.2962 \pm 0,0021$

Table 6: Statistical results of the dispersion relation fits ($\omega^2 = m \cdot k^2$).

The experimental data shows a high degree of correlation between ω^2 and k^2, with R^2 values consistently above $0,998$. This strict linearity confirms that the model $\omega^2 = (T/\rho)k^2$ accurately describes the system and proves that the string behaves as a non-dispersive medium, where the phase velocity remains constant regardless of the frequency mode.

Despite the excellent linear trend, the reduced chi-squared values are significantly high ($\chi^2_{red} > 3000$). This is not due to experimental error, but rather a consequence of the extreme precision of the stroboscope ($u(f) = 0,01$ Hz). When propagated to ω^2 values of 10^5, the resulting uncertainties are so small that the "ideal string" model cannot fully account for minor physical effects, such as the string's internal stiffness, which become detectable at this level of precision.

Finally, the calculated linear densities ($\rho \in [0,296,\ 0,\ 327]$ g/m) are highly consistent with the results from Section 2. The value obtained at $1,000$ N ($0,327 \pm 0,003$ g/m) is particularly accurate, deviating by less than 1% from the theoretical $0,33$ g/m. These results validate the dispersion relation as a robust method for characterizing the string, while the slight variations between tensions likely stem from minor systematic biases in the dynamometer or the effective length of the string.

4 Conclusion

The experimental analysis successfully validated the law of frequencies governing transverse standing waves in a bounded medium [1]. The least-squares regressions performed for all tension levels ($T = 1,000$ N, $0,900$ N, and $0,800$ N) returned coefficients of determination very close to unity ($R^2 > 0,998$). This strict linearity confirms that the string behaves as a non-dispersive medium within the studied frequency range, proving that the dispersion parameter α is negligible ($\alpha \approx 0$).

While the adjustments showed excellent structural trends, the reduced chi-squared values (χ^2_{red}) were significantly high. This does not indicate poor data collection, but rather reflects

the high instrumental precision of the digital stroboscope. The resulting error bars on the Y-axis are so narrow that the simplified "ideal string" model cannot fully absorb minor physical anomalies like internal string stiffness.

Finally, the linear density (ρ) of the string was successfully determined using both the individual dispersion plots and the global f_n/n vs $T^{1/2}$ representation. Both methods showed outstanding accuracy, deviating by less than 1% from the manufacturer's nominal value ($\rho_{theo} = 0,33$ g/m) [1]. Incorporating the dynamometer's uncertainty ($u_C(T) = 0,012$ N) proved essential to delivering a realistic and mathematically rigorous final uncertainty for the string's density.

5 Appendix

5.1 Graphics and fittings

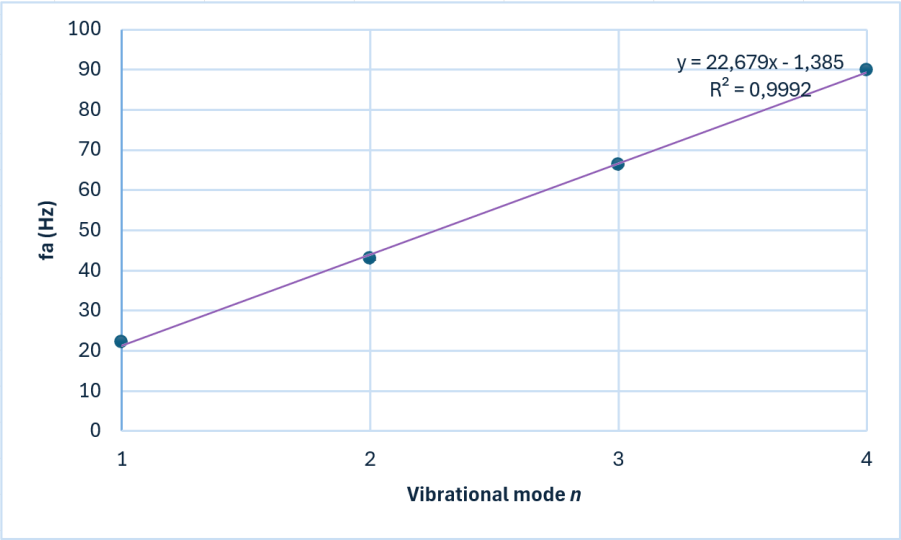


Figure 2: Linear fit of frequency f_a as a function of mode n for a string tension of 1.000 ± 0.012 N.

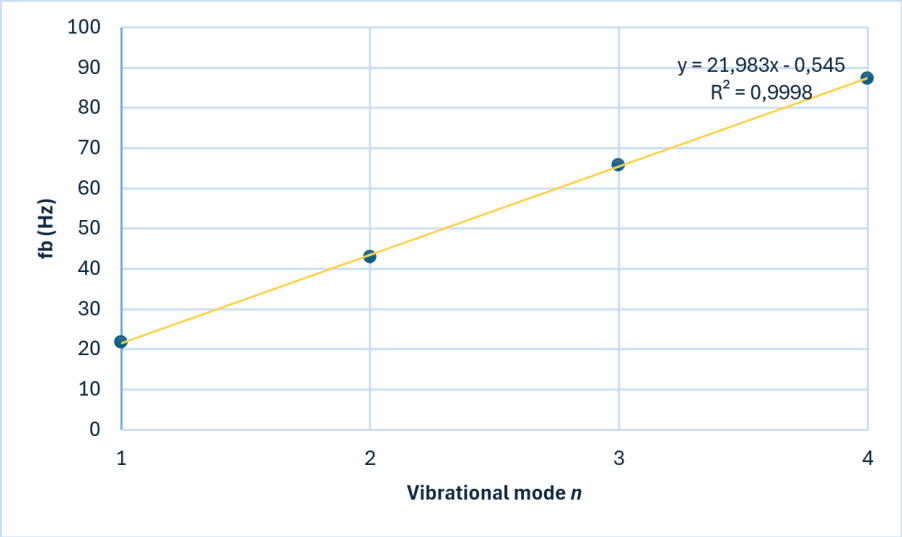


Figure 3: Frequency f_b versus mode number n for a tension of 0.900 ± 0.012 N.

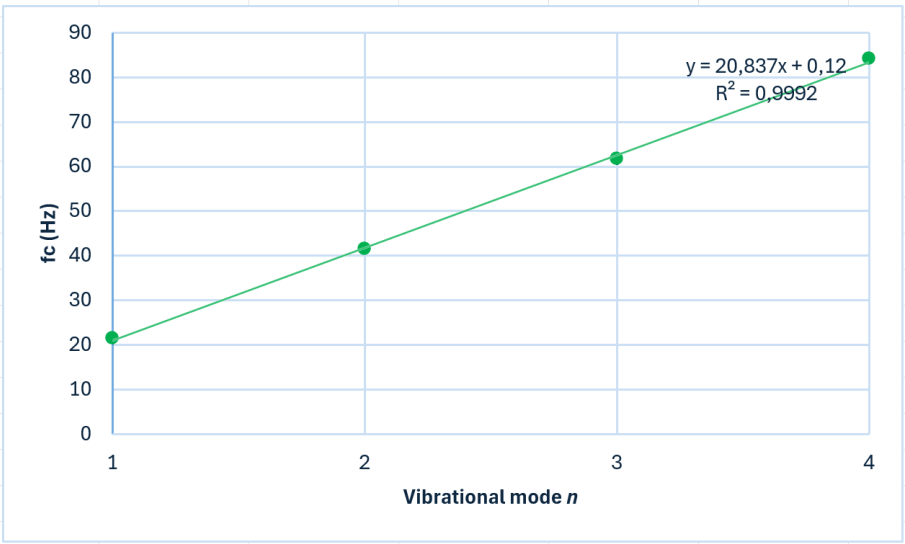


Figure 4: Linear relationship between frequency f_c and vibrational mode n at a tension of 0.800 ± 0.012 N.

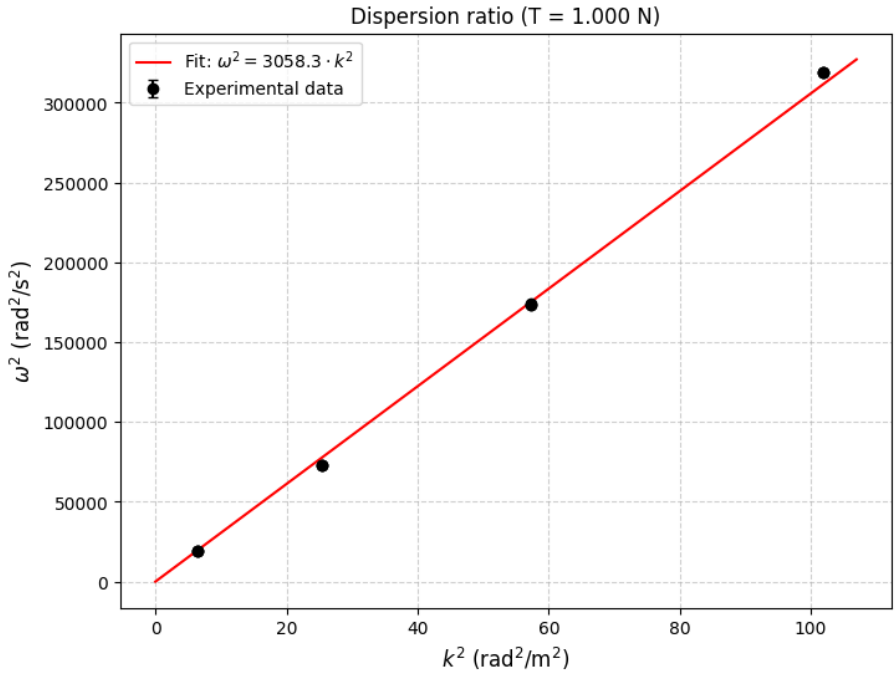


Figure 5: Linear fit of the dispersion relation $\omega^2 = f(k^2)$ at $T = 1.000$ N

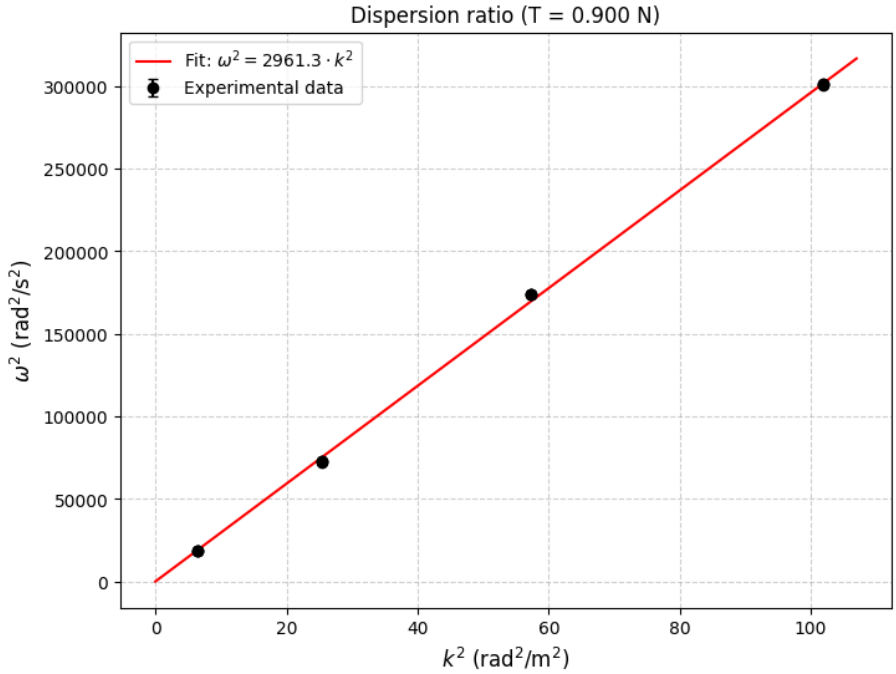


Figure 6: Linear fit of the dispersion relation $\omega^2 = f(k^2)$ at $T = 0{,}900$ N

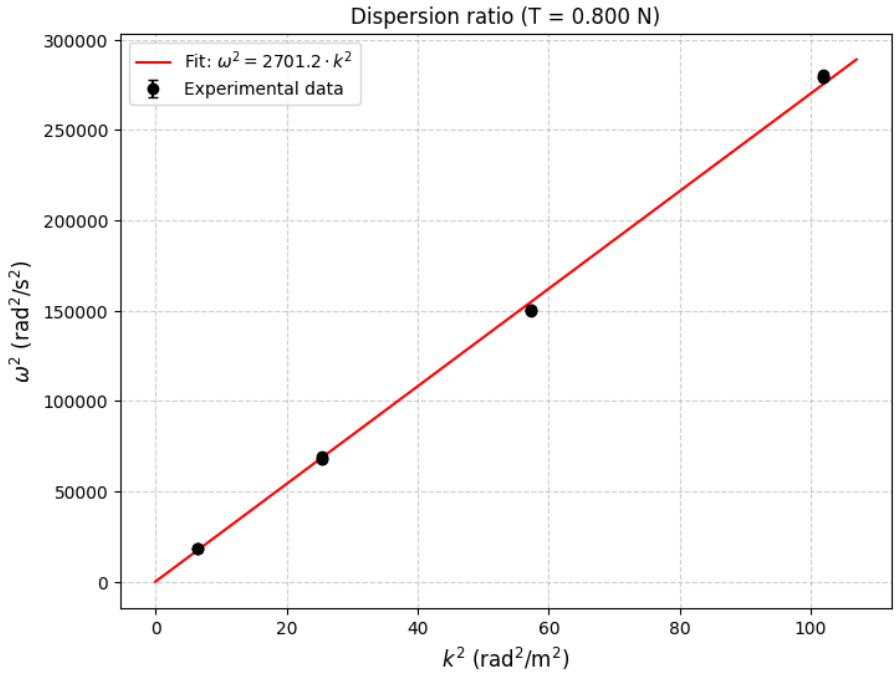


Figure 7: Linear fit of the dispersion relation $\omega^2 = f(k^2)$ at $T = 0{,}800$ N

5.2 Data tables

λ (m)	f_a (Hz)	f_b (Hz)	f_c (Hz)
2,4900 ± 0,0023	22,100 ±0,010	21,667 ± 0,010	21,417 ± 0,010
2,4900 ±0,0023	22,083 ± 0,010	21,683 ± 0,010	21,400 ± 0,010
2,4900 ± 0,0023	22,100 ± 0,010	21,683 ± 0,010	21,417 ± 0,010
1,2450 ± 0,0012	42,950 ± 0,010	42,883 ± 0,010	41,383 ± 0,010
1,2450 ±0,0012	42,933 ± 0,010	42,850 ± 0,010	41,867± 0,010
1,2450 ± 0,0012	42,933 ±0,010	42,867 ± 0,010	41,550 ± 0,010
0,83000 ± 0,00077	66,350 ± 0,010	66,367 ± 0,010	61,667 ± 0,010
0,83000 ±0,00077	66,300 ± 0,010	66,333 ±0,010	61,633 ± 0,010
0,83000 ± 0,00077	66,333 ± 0,010	64,633 ± 0,010	61,683 ± 0,010
0,62250 ±0,00058	89,933 ± 0,010	87,367 ± 0,010	84,117 ±0,010
0,62250 ±0,00058	89,883 ± 0,010	87,250 ± 0,010	84,100 ± 0,010
0,62250 ± 0,00058	89,867 ± 0,010	87,333 ± 0,010	84,317 ± 0,010

Table 7: Wavelength λ and frequencies f_a, f_b and f_c for each vibrational mode, with their corresponding uncertainties.

Mode (n)	$k^2(rad^2/m^2)$	$\omega^2(rad^2/s^2)$
1	$6,3685 \pm 0,0059$	$19280 \pm 17,50$
1	$6,3685 \pm 0,0059$	19245 ± 17
1	$6,3685 \pm 0,0059$	19280 ± 18
2	$25,474 \pm 0,024$	72837 ± 34
2	$25,474 \pm 0,024$	72769 ± 34
2	$25,474 \pm 0,024$	72769 ± 34
3	$57,317 \pm 0,053$	173801 ± 52
3	$57,317 \pm 0,053$	173540 ± 52
3	$57,317 \pm 0,053$	173697 ± 52
4	$101,896 \pm 0,095$	319287 ± 71
4	$101,896 \pm 0,095$	318932 ± 71
4	$101,896 \pm 0,095$	318861 ± 71

Table 8: Processed data for the dispersion relation fit at $T = 1,000$ N.

Mode (n)	$k^2(rad^2/m^2)$	$\omega^2(rad^2/s^2)$
1	$6,3685 \pm 0,0059$	18536 ± 17
1	$6,3685 \pm 0,0059$	18553 ± 17
1	$6,3685 \pm 0,0059$	18553 ± 17
2	$25,474 \pm 0,024$	72600 ± 34
2	$25,474 \pm 0,024$	72498 ± 34
2	$25,474 \pm 0,024$	72566 ± 34
3	$57,317 \pm 0,053$	173906 ± 52
3	$57,317 \pm 0,053$	173697 ± 52
3	$57,317 \pm 0,053$	173801 ± 52
4	$101,896 \pm 0,095$	301097 ± 69
4	$101,896 \pm 0,095$	300270 ± 69
4	$101,896 \pm 0,095$	300822 ± 69

Table 9: Processed data for the dispersion relation fit at $T = 0,900$ N.

Mode (n)	$k^2(rad^2/m^2)$	$\omega^2(rad^2/s^2)$
1	$6,3685 \pm 0,0059$	18110 ± 17
1	$6,3685 \pm 0,0059$	18076 ± 17
1	$6,3685 \pm 0,0059$	18110 ± 17
2	$25,474 \pm 0,024$	67613 ± 33
2	$25,474 \pm 0,024$	69225 ± 33
2	$25,474 \pm 0,024$	68170 ± 33
3	$57,317 \pm 0,053$	150259 ± 49
3	$57,317 \pm 0,053$	150064 ± 49
3	$57,317 \pm 0,053$	150308 ± 49
4	$101,896 \pm 0,095$	279313 ± 66
4	$101,896 \pm 0,095$	277858 ± 66
4	$101,896 \pm 0,095$	279313 ± 66

Table 10: Processed data for the dispersion relation fit at $T = 0,800$ N.

5.3 Uncertainties

5.3.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{11}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{12}$$

denotes the instrument's resolution ($\delta_V = 0.01$ V for voltage measurements and $\delta_T = 0.001$ s for the photogate period measurements). The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{13}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{14}$$

Finally, the expanded uncertainty is defined as:

$$U_C(x) = k \cdot u_C(x), \tag{15}$$

where k represents the coverage factor. Although the GUM [2] allows for various approaches depending on the sample size, we have adopted a coverage factor of $k = 2$. This choice is justified by the large volume of data collected (approximately 15000 points per experiment). With such a significant sample size, the distribution of the results effectively converges to a normal distribution. In this context, a factor of $k = 2$ provides a confidence level of approximately 95% [3].

5.3.2 Direct measurements

The magnitudes studied in the laboratory were the angular frequency ω, the tension T and the string's length L. For these measurements, we used instruments of resolution $\delta_\omega = 1$ rpm, $\delta_T = 0,02$ N and $\delta_x = 1$ mm respectively. Therefore,

$$U_C(\omega) = 2 \cdot \frac{\delta_\omega}{\sqrt{12}} = 0,58 \;\; rpm = 0,061 \;\; rad/s \tag{16}$$

$$U_C(T) = 2 \cdot \frac{\delta_T}{\sqrt{12}} = 0,012 \;\; N, \tag{17}$$

$$U_C(L) = 2 \cdot \frac{\delta_x}{\sqrt{12}} = 0,00058 \;\; m. \tag{18}$$

5.3.3 Specific Uncertainty Calculations

The combined uncertainties for each indirect magnitude are shown below. To determine the expanded uncertainty, as previously mentioned, the value will simply be multiplied by a coverage factor of $k = 2$.

- **General average value $\overline{x}$.**

$$u_c(\overline{x}) = \frac{1}{N}\sqrt{u_c(x_1)^2 + u_c(x_2)^2 + \cdots + u_c(x_N)^2} \tag{19}$$

- **Frequency f**

$$u_C(f) = \frac{u_C(\omega)}{2\pi} \;\; Hz \tag{20}$$

- **Wavelength λ**

$$u_C(\lambda) = \frac{2}{n} u_C(L) \;\; m \tag{21}$$

- **Density ρ**

$$u_C(\rho) = \frac{T}{4L^2a^2}\sqrt{\left(\frac{u_C(T)}{T}\right)^2 + \left(\frac{u_C(L)}{L}\right)^2 + \left(\frac{u_C(a)}{a}\right)^2} \;\; kg/m \tag{22}$$

- **Parameter a**

$$u_C(a) = a \cdot \sqrt{\left(\frac{u_C(c_{exp})}{c_{exp}}\right)^2 + \left(\frac{u_C(c/a)}{(c/a)}\right)^2}(m) \tag{23}$$

- **Parameter $\sqrt{T}$**

$$u_C(\sqrt{T}) = \left|\frac{\partial\sqrt{T}}{\partial T}\right| u(T) = \frac{1}{2\sqrt{T}} u(T) \tag{24}$$

- **Parameter $\overline{f_n/n}$**
  For each individual measurement $y_i = f_i/n_i$, the uncertainty is propagated as $u_C(y_i) = u_C(f)/n_i$. When averaging $N = 12$ measurements for a given tension level, the total instrumental component u_B is calculated by promediating the variances:

$$u_B(y) = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\left(\frac{u(f)}{n_i}\right)^2} \tag{25}$$

On the other hand,

$$u_A(y) = \frac{\sigma_{n-1}}{\sqrt{N}} = \frac{1}{\sqrt{N}}\sqrt{\frac{\sum(y_i - \bar{y})^2}{N-1}} \tag{26}$$

Then,

$$u_C(y) = \sqrt{u_A^2 + u_B^2} \tag{27}$$

In our experimental data, it was observed that $u_A \gg u_B$. Consequently, the final uncertainty is dominated by the statistical dispersion, which reflects the precision limit imposed by the visual detection of the string's maximum amplitude rather than the instrument's digital resolution.

- **Linear Density (ρ) (section 2.3)**

$$u_C(\rho) = \rho\sqrt{\left(2\frac{u_C(m)}{m}\right)^2 + \left(2\frac{u_C(L)}{L}\right)^2} \tag{28}$$

- **Wave number (k)**

$$u_C(k^2) = \left|\frac{\partial(k^2)}{\partial L}\right| u_C(L) = \left|-2\frac{n^2\pi^2}{L^3}\right| u_C(L) = 2 \cdot k^2 \cdot \frac{u_C(L)}{L} \tag{29}$$

- **Angular frequency (ω^2)**

$$u_C(\omega^2) = 2 \cdot (2\pi f) \cdot (2\pi \cdot u_C(f)) = 8\pi^2 f \cdot u_C(f) \tag{30}$$

- **Linear Density (ρ) (section 2.4)**

$$u_C(\rho) = \rho\sqrt{\left(\frac{u_C(T)}{T}\right)^2 + \left(\frac{u_C(m)}{m}\right)^2} \tag{31}$$

References

[1] Department of Applied Physics; University of Granada. *M9: Ondas estacionarias en una cuerda*. 2025.

[2] Turismo y Comercio Ministerio de Industria. *Guía para la Expresión de la Incertidumbre de Medida*. Páginas 37 y 38, Año 2008.

[3] Department of Applied Physics; University of Granada. *Notes for the Calculation of Uncertainties in Laboratory Sessions*. PRADO, 2025.